Information Technology University



Lab Manual

for

CS220: Artificial Intelligence

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Lab Task: Polynomial Curve Fitting Using SSE and Regularized Error

Objective

To fit a polynomial to a given dataset using two approaches:

1. **Minimizing the sum of squared errors (SSE).**
2. **Minimizing the regularized error.**

Instructions

1. **Use the provided dataset of 50 points (x, y).**
2. **Use polynomial degrees 2, 4, and 6 for fitting.**
3. **Compare the performance of the two methods using Mean Squared Error (MSE).**
4. **Implement gradient descent to update the weights vector `w`.**

DATASET:

x	y
-5	-123.35
-4.8	-111.29
-4.6	-99.25
-4.4	-88.9
-4.2	-78.84
-4	-72.15
-3.8	-63.79
-3.6	-55.63
-3.4	-48.76
-3.2	-38.5
-3	-33.16
-2.8	-24.85
-2.6	-18.2
-2.4	-12.15
-2.2	-7.53
-2	-3.91
-1.8	0.94
-1.6	3.82
-1.4	8.09
-1.2	11.18
-1	14.25
-0.8	16.43
-0.6	17.66
-0.4	18.19

-0.2	18.15
0	18.05
0.2	17.94
0.4	18.18
0.6	18.75
0.8	20.14
1	22.14
1.2	25.02
1.4	27.69
1.6	31.73
1.8	36.04
2	40.22
2.2	44.53
2.4	48.86
2.6	53.27
2.8	58.72
3	63.5
3.2	68.63
3.4	73.8
3.6	79.56
3.8	85.36
4	91.21
4.2	97.47
4.4	103.76
4.6	110.92
4.8	117.98
5	125.21

Part 1: Sum of Squared Error

1. Error Function

$$E(w) = \frac{1}{2}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

where $\hat{y}_i = \sum_{j=0}^{M} w_j x_i^j$, and M is the degree of the polynomial.

2. Weight Update Rule

Gradient of *E(w)* with respect to *wj*:

$$\frac{\partial E(w)}{\partial w_j} = -\sum_{i=1}^{N}(y_i - \hat{y}_i)x_i^j$$

Update rule:

$$w_j^{(t+1)} = w_j^{(t)} - \eta\frac{\partial E(w)}{\partial w_j}$$

Where η is the learning rate.

Part 2: Regularized Error

1. **Regularized Error Function**

$$E_{\text{reg}}(w) = \frac{1}{2}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 + \frac{\lambda}{2}\sum_{j=1}^{M}w_j^2$$

where λ is the regularization parameter.

2. Weight Update Rule

Gradient of *Ereg(w)* with respect to *wj*:

$$\frac{\partial E_{\text{reg}}(w)}{\partial w_j} = -\sum_{i=1}^{N}(y_i - \hat{y}_i)x_i^j + \lambda w_j$$

Update rule:

$$w_j^{(t+1)} = w_j^{(t)} - \eta\frac{\partial E_{\text{reg}}(w)}{\partial w_j}$$

Guidelines

1. **Data Preprocessing**
   Normalize *x* values to ensure numerical stability during gradient descent.
2. **Initialize Parameters**
   - Randomly initialize *w* vector for the chosen polynomial degree.
   - Choose a small learning rate *(η)* such as 0.01.
   - Set regularization parameter (λ) for Part 2 (e.g., *λ=0.1*).
3. **Iterative Update**
   - Compute the predicted values *y^i (y hat i)*.
   - Calculate the gradient for each *wj*.
   - Update *wj* using the respective update rules for Part 1 and Part 2.
   - Repeat until convergence (check for change in *w* or error below a threshold).
4. **Evaluation**
   Compute Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$